FINAL AGENDA

CHiMaD Phase Field Methods XVII

November 5 – 7, 2024

CHiMaD HQ @ NU: Hogan Suite 1160, 2205 Tech Dr., Evanston, IL, 60208

Zoom: https://northwestern.zoom.us/j/93102161697

All times in CST

Tuesday, November 5

12:00PM – 01:00PM Welcome & Lunch

01:00PM – 01:40PM **Microscopically-informed phase field models of grain growth: benchmarks and applications**
Marco Salvalaglio \ *TU Dresden – Germany (Virtual)*

01:40PM – 02:20PM **Applications of phase-field modeling to polycrystalline materials**
Katsuyo Thornton \ *University of Michigan*

02:20PM – 03:00PM Group Photo & Coffee Break

03:00PM – 03:40PM **Grain restructuring due to dislocation accumulation and subsequent microstructural evolution in nuclear materials**
Larry Aagesen \ *Idaho National Laboratory*

03:40PM – 04:20PM **Determining materials parameters by integrating phase field with experiments**
Jin Zhang \ *Northwestern University*

04:20PM – 05:30PM Discussion: **Phase field methods for grain growth and Benchmarks**

05:30PM Meeting Reception & Dinner

Wednesday, November 6

08:30AM – 09:00AM Breakfast

09:00AM – 09:40AM **Fusion of phase-field simulation and in-situ observation by Bayesian data assimilation**
Akinori Yamanaka \ *Tokyo University of Agriculture and Technology – Japan (Virtual)*

09:40AM – 10:10PM Lightning Talks (for speakers in Europe)
Triple junction benchmark combining capillary and bulk driving forces
Simon Daubner \ *Imperial College London – UK (Virtual)*

10:10AM – 10:40AM Coffee Break

10:40AM – 11:20AM **Interfacial energy minimization mechanisms during grain growth**
Michael Tonks \ *University of Florida*

11:20AM – 12:00PM **Dislocation microstructure considerations in phase field modeling of recrystallization**
Anter EI-Azab \ *Purdue University*

12:00PM – 01:00PM Lunch

01:00AM – 01:40PM **Parametrizing phase-field models for microstructure evolution with AMMBER**
Beck Andrews \ *University of Michigan*

01:40PM – 02:40PM **PFHub: Updates**
Daniel Wheeler \ *NIST*

02:40PM – 03:00PM **Best practices guide as a Jupyter book**
Mythreyi Ramesh \ *Northwestern University*

03:00PM – 03:30PM Coffee Break

03:30PM – *open ended* Lightning Talks

Matrix-free methods for solidification with melt convection in PRISMS-PF
Jason Landini \ *University of Michigan*

Phase field crystal study of deformation assistant nucleation and precipitation
Duncan Burns \ *Northwestern University*

Swift, a MOOSE based GPU accelerated spectral solver framework
Daniel Schwen \ *Idaho National Laboratory*

Phase-field crystal modelling of lamellar precipitation reactions
Alex Mamaev \ *McGill University – Canada*

Impact of secondary-phase particle morphology on grain growth in pure copper using a phase field simulation approach
M Nabil Bhuiyan \ *University of Connecticut*

Benchmarking phase-field problems with Fouriera
Bo Wang \ *Lawrence Livermore National Laboratory*

Applying Phase Field Method to Investigate Microstructural Evolution of an Irradiated UO2 Nuclear Fuel
Md Ali Muntaha \ *Purdue University*

Limits on the grain boundary energy vs. misorientation relationship in an orientation-field model
Philip Staublin \ *Northwestern University*

05:30PM Dinner

Thursday, November 7

08:30AM – 09:00AM Breakfast

09:00AM – 9:40AM **Formulation and testing of phase-field models for grain growth with inclination dependence**
Nele Moelans \ *KU Leuven – Belgium (Virtual)*

09:40AM – 11:00AM **Recommended Practices Guide**
Jon Guyer \ *NIST*

11:00AM – 12:00PM Discussion: **Next Steps**

12:00PM *Adjourn* & Lunch to-go